

Submit the following five exercises. You are expected to provide full solutions in a linear and coherent fashion. You are encouraged to work in groups, but your work must be your own and must be self-contained.

**Exercise 1.**

Find the rate of change of the area of a square with respect to the length of its side  $s$ , when  $s = 5$ .

**Exercise 2.**

Find the rate of change of the volume of a cylinder with respect to its radius, when its height is equal to its radius.

**Exercise 3.**

Recall that the volume and surface area of a sphere of radius  $r$  are given by  $V = \frac{4}{3}\pi r^3$  and  $A = 4\pi r^2$ , respectively. A spherical snowball is melting, and its surface area is decreasing at a rate of 0.5 centimeters squared per minute. How fast is the volume decreasing when the radius is 6 centimeters?

**Exercise 4.**

A boat is traveling on a river along the curve  $y = \sqrt{x+4}$ . When the boat is at the point  $(-1/2, \sqrt{7/2})$ , the  $y$ -coordinate of its position is decreasing at a rate of 10 knots. How fast is the  $x$ -coordinate of the boat changing at the point  $(-1/2, \sqrt{7/2})$ ? Explain which is changing faster, draw a picture of the situation, and explain why you would have expected this at the start of the problem.

**Exercise 5.**

A rocket travels vertically at a speed of 1200 kilometers per hour. An observer located 16 kilometers from the launching pad is tracking the rocket using a telescope. Assuming the Earth is flat, find the rate at which the angle between the telescope and the ground is increasing 3 minutes after lift-off.